Title: Dimensionality Reduction Methods: Toward Class Imbalance Handling in Financial Fraud Detection

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Authors: W. Banerjee, W. Haddad (Institute for Advanced Computing Research)

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Abstract:

Class imbalance handling remains a critical challenge for exploratory data analysis, particularly in financial fraud detection. This paper proposes a framework built on dimensionality reduction methods that explicitly accounts for causal effect estimation from observational data. We evaluate the framework on a ten-year retail transaction log and compare it against 3 baseline methods. Experimental results show that the proposed approach improves silhouette coefficient by 26.7% while keeping AUC-ROC within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of exploratory data analysis has changed how researchers and practitioners approach class imbalance handling, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, dimensionality reduction methods have attracted sustained attention as a promising route toward solving class imbalance handling in realistic settings such as financial fraud detection. Despite considerable progress, existing solutions for class imbalance handling tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in forecast MAPE frequently come at the expense of silhouette coefficient, forcing designers to accept trade-offs that are poorly understood.

This paper addresses these gaps by studying how dimensionality reduction methods interact with Bayesian hierarchical models when applied to class imbalance handling. Our work is motivated by the growing demand for financial fraud detection, where reliability and transparency requirements are especially strict. Understanding causal effect estimation from observational data is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Petrov et al. (2015) introduced an early framework based on dimensionality reduction methods, reporting a 19.2% improvement in forecast MAPE over previous baselines. Subsequent work by Takahashi et al. (2018) extended this line of research to financial fraud detection, although the evaluation was restricted to small-scale settings. A survey by Kim et al. (2011) categorised existing approaches to class imbalance handling into three families and highlighted the lack of standard benchmarks. Researchers working on exploratory data analysis have explored Bayesian hierarchical models as an alternative, with Chen et al. (2015) noting that the approach scales poorly beyond 32 components. Fernandez et al. (2012) analysed causal effect estimation from observational data and showed that purely heuristic solutions cannot guarantee consistent RMSE.

Several studies have criticised the reliance on synthetic workloads, arguing that results on a ten-year retail transaction log may not generalise to financial fraud detection. In contrast to these efforts, our work treats class imbalance handling and causal effect estimation from observational data jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to class imbalance handling as a persistent obstacle for financial fraud detection. Despite this body of literature, no prior study has systematically

compared dimensionality reduction methods and Bayesian hierarchical models under identical experimental conditions.

3. Methodology

We formalise class imbalance handling as a constrained optimisation task in which the objective is to maximise silhouette coefficient while keeping AUC-ROC within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on dimensionality reduction methods, and a validation layer that incorporates Bayesian hierarchical models. All experiments were conducted on a ten-year retail transaction log, which provides 8 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 20 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies dimensionality reduction methods to reduce the effect of causal effect estimation from observational data, followed by a calibration step that stabilises silhouette coefficient. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 3 baseline methods, including a conventional approach built on Bayesian hierarchical models and two recent variants reported in the literature.

The proposed design improves silhouette coefficient by 26.7% relative to the strongest baseline and reduces RMSE by 13.7%, a statistically significant difference ($p < 0.05$). Ablation experiments show that removing the Bayesian hierarchical models component reduces silhouette coefficient by approximately 9.5%, confirming its contribution. A sensitivity analysis over 128 parameter settings shows that performance remains stable, with variation in AUC-ROC below 6.6%.

4. Conclusion

This paper investigated class imbalance handling through the lens of dimensionality reduction methods, and demonstrated that a carefully designed pipeline can improve silhouette coefficient by 26.7% on a ten-year retail transaction log. The findings suggest that exploratory data analysis will continue to benefit from closer integration of dimensionality reduction methods and Bayesian hierarchical models, particularly in financial fraud detection. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of causal effect estimation from observational data. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for class imbalance handling.